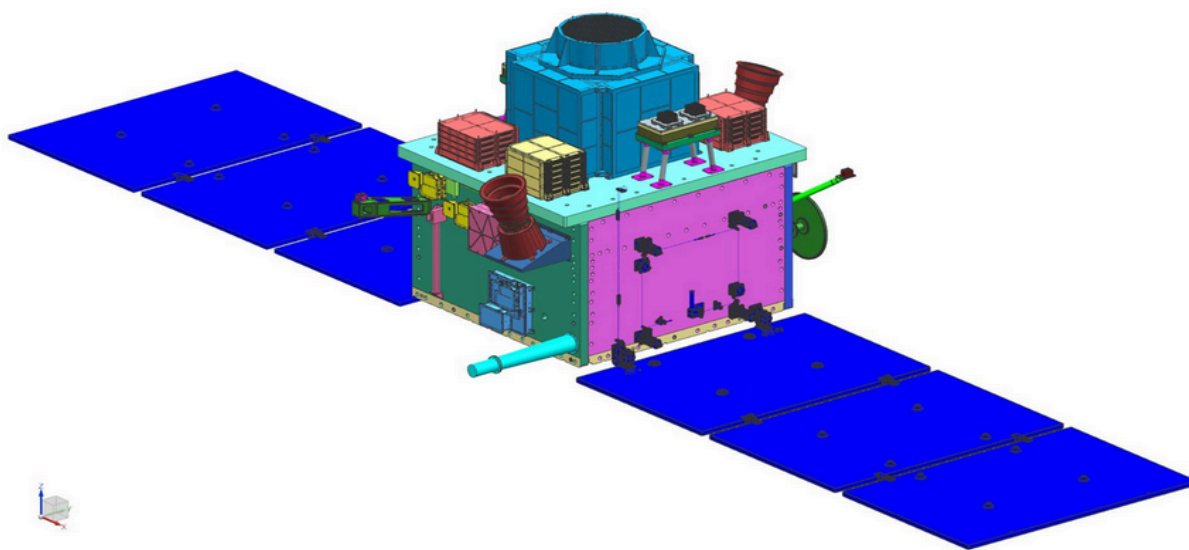


XPoSat-POLIX

User Handbook



User Guide for Scientific Data Analysis
Level2
V1.0, October 2025

Document Revision History

Prepared By	Nandini Anand, Ketan Rikame, Kinjal Roy	
Reviewed by	Vikram Rana, Rishin P V	
Approved by	Biswajit Paul	

Document Change History

Version	A/M/D	Description
V1.0	--	First version

A : Addition; M : Modification; D : Deletion

Contents

1. Introduction
2. POLIX data analysis method
 - 2.1 Source, Background, and Earth Occultation Durations
 - 2.2 Level2 Pipeline Input Data Description
 - 2.3 Reference Vector and Azimuthal Angle
 - 2.4 Azimuthal Counts and Exposure Distribution
 - 2.5 Exposure-Weighted Modulation Curve
3. Overview of POLIX Data and Pipeline
 - 3.1 Data Overview
 - 3.2 Pipeline Overview
4. Downloading Data and File Content
 - 4.1 Naming Conventions
 - 4.2 Level1 Data Files
 - 4.3 Level2 Data Files
5. Installation and Setup
 - 5.1 System Requirements
 - 5.2 Dependencies: Core Software
 - 5.3 Python Environment
 - 5.4 Download and Installation
 - 5.5 Pipeline Demonstration
6. POLIX Level2 Software
 - 6.1 polix_l2_lcpha
 - 6.2 polix_l2_source_azimuth
 - 6.3 polix_l2_exposure_azimuth
 - 6.4 polix_l2_EnergyRes_Shifted_Src_Azimuth
 - 6.5 polix_l2_Weighted_Exposure
7. User Parameters
8. Calibration Data & Associated Files
 - 8.1 Channel to energy conversion
9. Barycenter correction
10. Pipeline Walkthrough
 - 10.1 Default Run of pipeline
 - 10.2 Pipeline run with user inputs
 - 10.3 Running individual POLIX Modules
 - 10.3.1 Lightcurve and spectrum module
 - 10.3.2 Source azimuth module
 - 10.3.3 Exposure module
 - 10.3.4 Energy dependent source azimuth module

10.3.5 Weighted exposure module

11. Timeline of POLIX detectors

12. Variability of background modulation and limitations

A Annexure

A1 Fits File Description

A1.1 L1 products

A1.2 L2 products

List of figures

1. POLIX Instrument
2. Reference Vector (SxZ) and Azimuthal Angle (θ)
3. A Sample Azimuthal Modulation Curve
4. Overview of POLIX Level2 data analysis pipeline work flow
5. The inputs for the default run of the POLIX pipeline
6. Pipeline parameter confirmation
7. The last message that is shown at the end of the pipeline run
8. Lightcurve of the source covering time with 100 second bin size for Proposal ID: G01_0003
9. Spectrum of the source observation duration for Proposal ID: G01_0003
10. The azimuthal distribution of the count rate for Proposal ID: G01_0003
11. The inputs used for a custom run of the POLIX pipeline
12. Custom selections are shown while running the pipeline with multiple user input files
13. A screenshot of a sample user GTI file
14. A sample userdetsidepha.txt file
15. An example output for the count lightcurve creation module
16. An example output for the event lightcurve and spectrum creation module
17. Content of the input text file for polix-lc-count-creation.v1.0 module
18. An example output for the source azimuth module
19. An example output for the exposure module
20. An example output for the energy dependent source azimuth module

List of Tables

1. Summary of modules in POLIX Level2 pipeline and their functionality
2. L1 and L2 data products along with their identifiers in the product name
3. Criteria for Source, Background and Earth Occultation observation durations
4. Description of L1 data product: Event File
5. Description of L1 data product: Count File
6. Description of L1 data product: Final_GTI File
7. Description of L1 data product: Science_GTI File
8. Description of L1 data product: Housekeeping File
9. Description of L1 data product: Attitude file
10. Description of L2 data product: Light Curve
11. Description of L2 data product: Spectrum
12. Description of L2 data product: Exp_Azimuth_Roll_L2.fits
13. Description of L2 data product: Exp_Azimuth_Yaw_L2.fits
14. Description of L2 data product: Src_Azimuth_Roll_L2.fits
15. Description of L2 data product: Src_Azimuth_Yaw_L2.fits
16. Description of L2 data product: EnergyRes_Src_Azimuth_Roll_L2.fits
17. Description of L2 data product: WeightedRoll_L2.fits

1 Introduction

The Indian X-ray Polarimeter (POLIX) developed by the Raman Research Institute (RRI) in collaboration with the U R Rao Satellite Centre (URSC) is the primary scientific payload onboard India's X-ray Polarimeter Satellite (XPoSat). POLIX is designed to measure the degree and angle of polarization of X-rays in the medium energy range of 8-30 keV from various bright astronomical sources, including black holes, neutron stars and pulsar wind nebulae.

The instrument consists of a collimator, a low atomic mass scatterer, and four X-ray proportional counter detectors surrounding the scatterer. Incoming polarized X-rays undergo anisotropic Thomson scattering within the scatterer, and the detectors measure the resulting distribution of source photons along with various components of background, which provides information about the polarization of the X-rays. The collimator restricts the field of view to 3 degrees by 3 degrees, ensuring that only one bright source is observed at a time. POLIX commenced its scientific observations in March 2024.

POLIX data reduction and analysis requires dedicated software tools. The POLIX software is a dedicated analysis tool designed for processing and interpreting data from POLIX. The software facilitates data calibration, event processing, and generation of azimuthal intensity modulation curves from observations of the astrophysical sources. The POLIX software is designed to enable users to process detector data and extract the degree and angle of polarization using specialized algorithms. The software is designed to support processing of archival data, ensuring accurate and reliable scientific results. For a more detailed description of the instrument, see Section 2.

This document provides an overview of the POLIX software, its core functionalities, and limitations in X-ray polarisation measurement due to a variable background modulation.

In the presence of a variable background modulation, and that of the spikes in some observations, X-ray polarization measurement is not possible with the currently released data, in which a source is observed for a few days and multiple blank sky regions are observed for additional days. A new strategy of observing a bright X-ray source and nearby blank sky regions has been adopted for polarization measurements with POLIX. Data from these observations and methods for background subtraction will be detailed in future data/handbook release. See section 11 for more details.

2 POLIX data analysis method

The primary objective of the POLIX Level2 PIPELINE is to generate exposure-weighted azimuthal modulation curves of detected X-ray photons, from which polarization information can be derived.

POLIX is a Thomson scattering polarimeter. The collimator restricts the field of view to $3^\circ \times 3^\circ$. Incoming X-ray photons are scattered by a central beryllium scatterer and detected by four position-sensitive proportional counters arranged around it. Each detector has 12 anode cells which are divided into two channels, with the odd and even anode wires connected in series to make a channel. In total, there are 48 anodes (4 detectors $\times$ 2 channels $\times$ 6 anodes each), distributed in the azimuthal plane. During observations, the satellite rotates around its Pitch-axis at ~ 0.2 rpm, such that every anode samples the full 360° azimuthal range. In principle, each anode acts as an independent polarimeter.

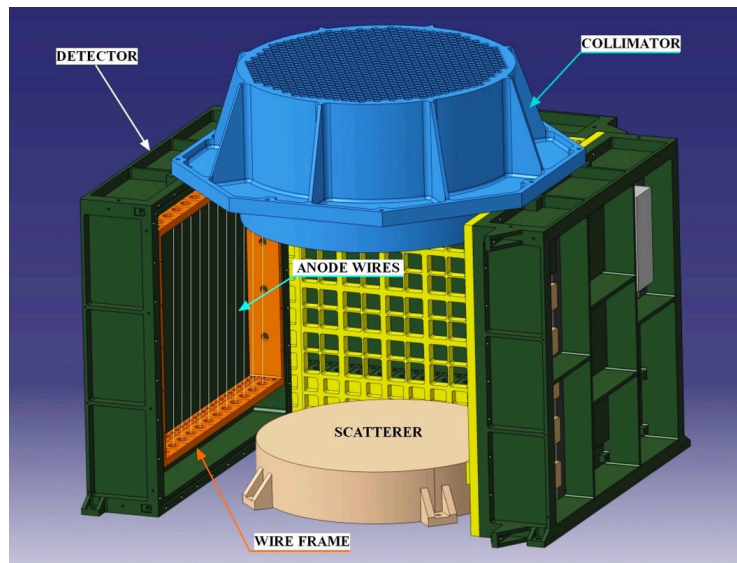


Figure 1: POLIX Instrument

2.1 Source, Background, and Earth Occultation Durations

POLIX operates in a low-Earth equatorial orbit with an inclination of about six degrees. For data analysis, the observation duration is divided into three categories: source, background, and Earth occultation durations. Please note that these are only names of different types of observation

durations. As described below, the background and source durations have different pointings and the light curves and spectra extracted from the background observation duration cannot be used to subtract background from the light curves and spectra from the source observation duration.

Earth Occultation: If the source elevation is less than zero, the source is below the Earth's horizon and cannot be observed. The elevation is defined as the angle between the Pitch-axis and the Earth horizon tangent. These intervals are identified as Earth occultation durations.

Background: When the source elevation is greater than zero, but the angular offset between the Pitch-axis and the pointing direction of the source exceeds 0.2° , the source lies outside the flat-top response of the POLIX collimator and does not contribute optimally to the detected signal. These intervals are classified as background durations.

Source: When the elevation is greater than zero and the angular offset between the Pitch-axis and the pointing direction is less than 0.2° , the source lies well within the flat-top response of the POLIX collimator. These intervals are identified as source durations.

In addition to these conditions, source observations are further constrained by the South Atlantic Anomaly (SAA) durations.

2.2 Level2 Pipeline Input Data Description

The input to Level2 PIPELINE is the source event file, which contains information for each detected photon when the instrument was pointing at the desired sky coordinates. For every photon, the file records the detection time, the detector and channel number, the anode in which it was registered, the photon energy channel, and the satellite orientation at that moment (RA and DEC of the Roll and Yaw-axis of the spacecraft). The event file also includes the Good Time Intervals (GTIs), which specify the observation periods corresponding to the desired source coordinates. In addition, the orbit file (.att) provides the spacecraft orientation at regular time intervals and is essential for exposure estimation. All times in both the files are expressed in Universal Time (UT), measured as seconds elapsed since the reference epoch MJDREF (1 January 2023).

2.3 Reference Vector and Azimuthal Angle

In order to measure the azimuthal distribution of detected photons, we need to calculate their angle with respect to a fixed vector in the azimuthal plane. Since the instrument rotates during observation, we define this fixed vector in the sky plane as $S \times Z$, where S is the source pointing direction and Z is the north vector (0, 0, 1) in the equatorial coordinate system.

This cross product lies in the azimuthal plane, is perpendicular to the pointing direction, and remains fixed in the sky regardless of satellite position and rotation. For each detected photon, the PIPELINE calculates the angle of the Roll-axis relative to this reference vector (measured clockwise when looking from the pointing direction).

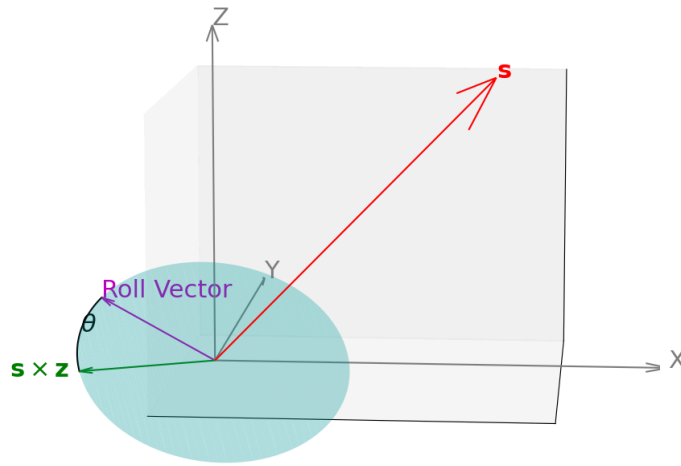


Figure 2: Reference Vector ($S \times Z$) and Azimuthal Angle (θ)

Each anode has a different position relative to the Roll-axis. For each photon detected in an anode cell, simulation-derived shift values (which also depend on photon energy) are added to the Roll-axis azimuthal angle at that instant to determine the azimuth angle of the corresponding anode cell relative to the reference vector (`/caldb/sshift_v4.txt`) at the time the photon is detected. In this way, the azimuthal angle of every detected photon is estimated consistently in the fixed sky reference frame.

2.4 Azimuthal Counts and Exposure Distribution

Since the shift factors depend on both the anode (48 in total) and the photon energy (8192 channels), the azimuthal photon count distribution as a function of the angle of the Roll-axis

from the reference vector is first calculated for every anode at all energy channels. The resulting product is a three-dimensional data cube of size 48 (anodes) $\times$ 8192 (energy channels) $\times$ 360 (azimuthal bins at 1° resolution). In this step, the count distribution with respect to the Roll-axis azimuthal angle is estimated separately for each anode and energy channel. Appropriate shifts are then applied so that, in the shifted azimuthal counts distribution, the azimuthal angle represents the same quantity for all the anodes: the angle of the detected photons relative to a fixed reference vector in sky coordinates.

Exposure in the azimuthal plane is calculated using the spacecraft orbit file (.att), which provides the Roll-axis position at 100 ms intervals, together with the Good Time Intervals (.GTI). When the Roll-axis moves across azimuthal angle bin boundaries, linear interpolation is used to estimate the time spent in each azimuthal angle bin. The resulting Roll-axis azimuthal exposure distribution is then shifted, in the same way as the counts distribution, to obtain the azimuthal exposure distribution corresponding to each anode.

2.5 Exposure-Weighted Modulation Curve

For each anode, the PIPELINE computes the azimuthal count rate by dividing the shifted photon counts distribution by the shifted exposure distribution. Finally, modulation curves from selected anodes and energy ranges are added to produce the overall azimuthal modulation curve, which is the key output for polarization measurement (See Figure 3).

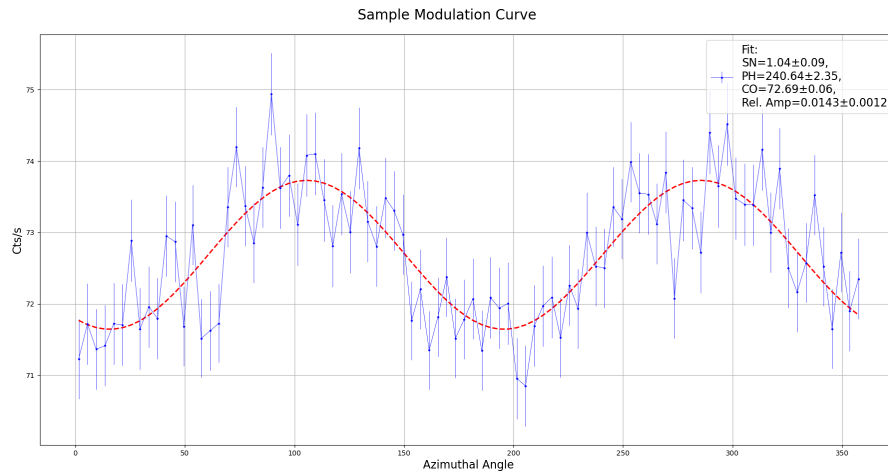


Figure 3: A Sample Azimuthal Modulation Curve

3 Overview of POLIX Data and Pipeline

3.1 Data Overview

The data received from the POLIX payload passes through a series of steps in multiple processing pipelines. Three major data levels have been designated for long-term storage. These are:

Level0

This is the raw telemetry data received from the satellite, accompanied by auxiliary data converted to suitable format for further processing at the POC. The science data from the POLIX payload is in units of data frames of size 2048 bytes in fits format and the remaining data is in the form of FITS files with time series of values. It is archived internally and is not made available for public use.

Level1

The primary content of Level1 data is event-wise information, providing all relevant details for each detected X-ray photon to support further scientific investigation. All auxiliary information required for subsequent processing is also collated at this level and stored together with the corresponding science data. In addition to the event-wise data from the detectors, additional count rate data is available with 100 ms binning.

Level2

This dataset contains standard science products derived from Level1 data. Level2 data, in FITS format, is also available for scientific use.

Level0 data is obtained from the ISSDC (Indian Space Science Data Centre) archived and processed at the POC on a day-by-day basis to generate Level1 products. The Level1 data is later merged according to proposal ID, and the merged products, along with the corresponding Level2 data, are archived back at the ISSDC.

The directory structure and FITS file formats of Level1 and Level2 data are described in Section 4.2 and 4.3.

3.2 Pipeline Overview

The software used to process data from Level1 to Level2 includes user-configurable parameters. At the Payload Operation Centre (POC), the pipeline is run with a default configuration to automatically generate the standard Level2 data products. Users, however, can run the same software with their own chosen parameters to create customized products. Please refer to section 7 for user parameter options. The Level2 pipeline software is publicly released for general use.

The processing workflow is implemented in Fortran using the CFITSIO library for FITS file handling, with additional processing and automation performed through Python scripts and shell scripts, executed on a Linux platform.

POLIX Level2 data reduction pipeline is organized into different modules for each specific task. In figure 4, work flow of the pipeline is shown. Table 1 summarizes the tasks performed by various individual modules.

Data reduction for POLIX is performed in three stages. Each of these stages involves execution of several tasks of the pipeline.

LEVEL0:

- Conversion of raw telemetry data to FITS format.
- Inclusion of all relevant auxiliary data.

LEVEL1:

The Level1 software is designed to process Level1 data for:

- Data format verification,
- Merging of multiple datasets from a single POC ID,
- Ordering the data frames based on frame counter number
- Removing duplicate dataframes, if any
- Create time calibration database,
- Separate bin counter data and event data,
- Removal of multiple simultaneous events,
- Convert system time to UT using the time calibration,

- Source overlap check using orbit file,
- Create time windows for different types of observations,
- Additional Time filtering and updating the Final GTI accordingly,
- Performing the gain correction of pha values and adding anode ID of each X-ray event,
- Splitting the event data and count data based on time windows for different types of observations,
- Determine the satellite aspect information and associate it with each X-ray event in the event data.

LEVEL2:

Extraction of scientific products:

- Light curves
- Spectra
- Modulation curves

All products are in FITS format. As Level0 and Level1 software are not available for public use, the following sections describe only the Level2 pipeline software, which is publicly accessible.

The pipeline tasks can either be executed individually or by using the **POLIXL2PIPELINEV1P1** tool, which allows the user to run required stages of the pipeline tasks together. It is advised that users use the **POLIXL2PIPELINEV1P1** module and use the individual modules only if it is required for specific analysis cases.

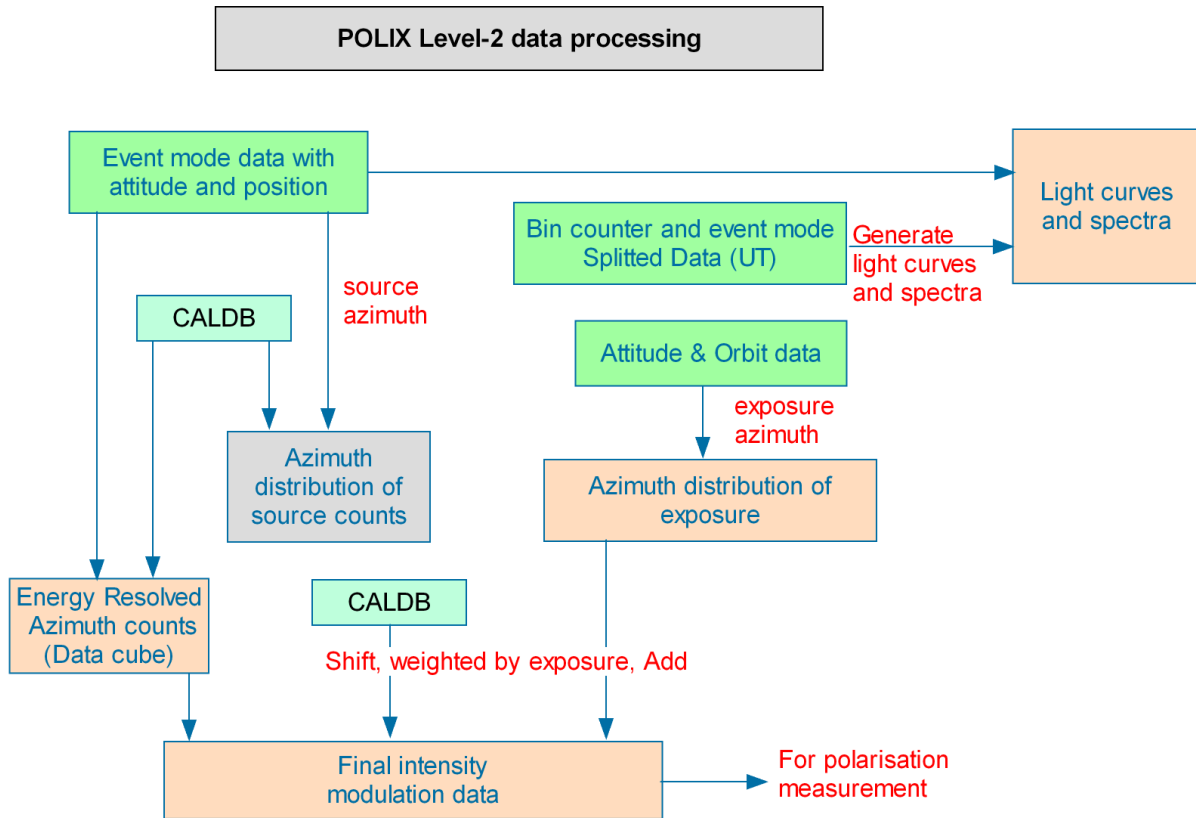


Figure 4: Overview of POLIX Level2 data analysis pipeline work flow.

The POLIX Level2 data processing pipeline begins with three primary inputs, shown in green in Figure 4, namely the event mode data with spacecraft attitude and position, the bin counter and event mode Splitted data in UT, and the attitude and orbit data. From these inputs, the splitted event data are used to generate light curves and spectra and count data to generate light curves, while the attitude and orbit data provide the azimuth distribution of exposure. In parallel, the event mode data are transformed into energy-resolved azimuth counts for each anode, forming a data cube. These intermediate products, highlighted in yellow, are then combined through shifting, weighing by exposure, and addition to produce the final azimuthal intensity modulation data, which is used to measure X-ray polarisation. An additional step, the azimuth distribution of source counts (shown in grey), is derived but not carried forward into the final processing chain.

Table 1: Summary of modules in POLIX Level2 pipeline and their functionality.

Sl.No.	Module Name	Functionality
1.	polix_l2_lcpha	To create light curves from both Binned Data and Events Data and spectrum for event data with options for various selections, including time interval selection.
2.	polix_l2_Source_Azimuth	To create azimuthal distribution of counts for the 48 cells.
3.	polix_l2_Exposure_Azimuth	To create azimuthal distribution of exposure in selected time intervals.
4.	polix_l2_EnergyRes_Shifted_Src_Azimuth	To create a data cube, for each anode cell, PHA resolved and 360 bins of Roll azimuthal angle.
5.	polix_l2_polarization	To create weighted, shifted modulation curves with Energy dependent Shift.

4 Downloading Data and File Content

POLIX data (Level1 and standard Level2 data products) is available for download from the ISSDC data archive. These Level1 and Level2 data products are packaged separately and provided in .tgz format.

Since Level1 data files can be large, the corresponding .tgz file is split into multiple parts (each ~1 GB) to facilitate easier transfer. The naming convention for the split Level1 files is as follows:

X01_POL_<Proposal-Id>_<YYYY>_L1_Vmpn_part<NNN>_Of_<MMM>.tgz

Here, the <NNN> represents the part number (always in three-digit format, e.g., 001, 002, 003...), and the <MMM> represents the total number of parts.

Users can re-merge these split Level1 files using the following commands:

On Linux (Ubuntu):

Open a terminal in the directory containing all .tgz parts and run:

```
cat $(ls -v X01_POL_<Proposal-Id>_<YYYY>_L1_Vmpn_part*.tgz) >
X01_POL_<Proposal-Id>_<YYYY>_L1_Vmpn.tgz
```

On Windows:

Open Command Prompt in the directory containing the parts and run:

```
copy /b X01_POL_<Proposal-Id>_<YYYY>_L1_Vmpn_part*.tgz
X01_POL_<Proposal-Id>_<YYYY>_L1_Vmpn.tgz
```

It is recommended that the merging of files larger than 4 GB be done in storage spaces which can support files of the final merged file size. External storage with FAT32 format cannot be used to merge files with final size greater than 4 GB, instead storage with exFAT and NTFS formats can be used.

4.1 Naming Conventions

Once downloaded (and L1 products re-merged as described above), products will be bundled in a compressed archive named

X01_POL_<Proposal-Id>_<YYYY>_<level>_Vmpn.tgz

The .tgz file can be extracted normally (e.g., with tar -xvzf on Linux or common archive tools like 7-Zip/WinRAR on Windows). On extraction, the directory will have the following name:

X01_PLX_<Proposal_ID>_000000_<level>_Vmpn

where,

<Proposal_ID> is an 8-character string indicating the unique proposal ID of the observation.

<Vmpn> refers to the data product version (e.g. V1p1).

<YYYY> refers to the year of observation.

<level> stands for the 2-character product level ('L1' or 'L2').

This folder name is typically 31 characters, and it contains the standard data products of the specified level. The products have the following naming convention:

X01_PLX_<proposal-ID>_000000_<filetype>_<level>.<ext>

where,

<ext> denotes extensions.

<filetype> denotes the type of product and summarized in Table 2

Table 2: L1 and L2 data products along with their identifiers in the product name.

File Description	Identifier	ext	Level
Event File	ProcessedSourceUTevents SourceUTevents BackgroundUTevents EarthoccultationUTevents	fits	Level1
Count Data File	SourceUTcounts BackgroundUTcounts EarthoccultationUTcounts	fits	Level1
GTI File	FinalUT_gti UTscience_gti	gti	Level1
Attitude file	AUX	att	Level1

HouseKeeping File	HK	hk	Level1
Light curve	ProcessedEventdataSource_Det1/2/3/4 EventdataSource EventdataBackground EventdataEarthoccultation CountdataSource CountdataBackground CountdataEarthoccultation	lc	Level2
Spectrum	ProcessedEventdataSource_Det1/2/3/4 EventdataSource EventdataBackground EventdataEarthoccultation	pha	Level2
Exposure Azimuth	Exp_Azimuth_Roll/Yaw	fits	Level2
Source Azimuth	Src_Azimuth_Roll	fits	Level2
Data cube	EnergyRes_Src_Azimuth_Roll	fits	Level2

4.2 Level1 Data Files

Level1 data is organized by proposal ID, with data from multiple observation days belonging to the same proposal ID merged into a single dataset. Each directory contains the corresponding processed event files, Good Time Interval (GTI) file based on various criteria, and count data for that proposal. Content of the POLIX level1 compressed file is shown below as an example and the details of each of the files is listed after that.

Untar **X01_PLX_G01_0003_000000_L1_V1p1.tar.gz**

```

└── X01_PLX_G01_0003_000000_L1_V1p1
    ├── caldb
    │   ├── Anode-Identifier.fits
    │   ├── G01_0003_000001_gain_para_normalized.txt
    │   └── sshift_v4.txt
    ├── hk
    │   ├── X01_PLX_G01_0003_000000_AUX_L0_V1p1.att
    │   └── X01_PLX_G01_0003_000000_HK_L0_V1p1.mkf

```

```

├── InputTxt
│   ├── det1lcpha.txt
│   ├── det2lcpha.txt
│   ├── det3lcpha.txt
│   ├── det4lcpha.txt
│   ├── det.txt
│   ├── EnergyRes_Src_Azimuth.txt
│   ├── Exposure-azimuthal.txt
│   ├── l2_energyres_polarization_input.txt
│   ├── lc-bg-count.txt
│   ├── lc-eo-count.txt
│   ├── lcpha-bg-evnt.txt
│   ├── lcpha-eo-evnt.txt
│   ├── lcpha-sc-evnt.txt
│   ├── lcpha-sc-Processed-evnt.txt
│   ├── lc-src-count.txt
│   ├── sampleuserdetsidepha.txt
│   └── Source-azimuthal.txt
├── X01_PLX_G01_0003_000000_BackgroundUTcounts_L1.fits
├── X01_PLX_G01_0003_000000_BackgroundUTEvents_L1.fits
├── X01_PLX_G01_0003_000000_EarthoccultationUTcounts_L1.fits
├── X01_PLX_G01_0003_000000_EarthoccultationUTEvents_L1.fits
├── X01_PLX_G01_0003_000000_FinalUT_gti_L1.gti
├── X01_PLX_G01_0003_000000_ProcessedSourceUTEvents_L1.fits
├── X01_PLX_G01_0003_000000_SourceUTcounts_L1.fits
├── X01_PLX_G01_0003_000000_SourceUTEvents_L1.fits
└── X01_PLX_G01_0003_000000_UTscience_gti_L1.gti

```

1. **Science Data File (.fits):** Event files are generated separately for source, background, and Earth occultation time intervals, and contain information such as PHA1, PHA2, detector ID, and the anode ID. Count data, also organized for source, background, and Earth occultation intervals, include accepted counts, ULD1, ULD2, LLD1, LLD2, and anti-coincidence counts, recorded every 100 ms. A processed Event file which has additional information, anode identification and orientation of the satellite at the time of X-ray detection are required for creation of the modulation curve.

2. **Good Time Intervals (.GTI):** GTI files store the time intervals corresponding to source, background, and Earth occultation, HV for each detector, determined based on various selection criteria. Please Refer to Table 3. for source, background, and Earth occultation Criteria.
3. **MKF (.mkf):** This gives a time-tagged recording of 131 different health parameters of the POLIX.
4. **Attitude (.att):** Gives time-tagged satellite ephemeris and attitude information, in terms of RA, Dec values of the pointing directions of the three reference axes of the satellite and along with quality flag information.
5. **Caldb:** Refer Section 8.
6. **InputTxt:** Refer Section 7.

Table 3 : Criteria for Source, Background, and Earth Occultation observation durations.

Elevation Angle	Angular Offset	Condition / Value	Classification
< 0	-	Source below Earth's horizon	Earth Occultation
> 0	> 0.2°	Outside flat-top response of POLIX collimator	Background
> 0	< 0.2°	Within flat-top response of POLIX collimator	Source

4.3 Level2 Data Files

The Level2 data archive contains the final products, including light curves derived from count and event data, and spectra generated from event data. For processed events, separate light curve (LC) and PHA files are created for each detector to store the corresponding time-series and spectral information. The final modulation curve is produced through the default processing at the POLIX POC. The keyword `SOFT_VER = 'V1P0'` in the FITS header specifies the version of the software used to generate the products.

Content of the POLIX Level2 compressed file is shown below as an example.

```

└─ X01_PLX_G01_0003_000000_L2_V1p1
  └─ InputTxt
    └─ det1lcpha.txt
    └─ det2lcpha.txt
    └─ det3lcpha.txt
    └─ det4lcpha.txt
    └─ EnergyRes_Src_Azimuth.txt
    └─ Exposure-azimuthal.txt
    └─ l2_energyres_polarization_input.txt
    └─ lc-bg-count.txt
    └─ lc-eo-count.txt
    └─ lcpha-bg-evnt.txt
    └─ lcpha-eo-evnt.txt
    └─ lcpha-sc-evnt.txt
    └─ lcpha-sc-Processed-evnt.txt
    └─ lc-src-count.txt
    └─ sampleuserdetsidepha.txt
    └─ Source-azimuthal.txt
  └─ Polix_l2_lcpha
    └─ X01_PLX_G01_0003_000000_CountdataBackground_L2.lc
    └─ X01_PLX_G01_0003_000000_CountdataEarthoccultation_L2.lc
    └─ X01_PLX_G01_0003_000000_CountdataSource_L2.lc
    └─ X01_PLX_G01_0003_000000_EventdataBackground_L2.lc
    └─ X01_PLX_G01_0003_000000_EventdataBackground_L2.pha
    └─ X01_PLX_G01_0003_000000_EventdataEarthoccultation_L2.lc
    └─ X01_PLX_G01_0003_000000_EventdataEarthoccultation_L2.pha
    └─ X01_PLX_G01_0003_000000_EventdataSource_L2.lc
    └─ X01_PLX_G01_0003_000000_EventdataSource_L2.pha
  └─ Polix_l2_polarization
    └─ X01_PLX_G01_0003_000000_EnergyRes_Src_Azimuth_Roll_L2.fits
    └─ X01_PLX_G01_0003_000000_Exp_Azimuth_Roll_L2.fits
    └─ X01_PLX_G01_0003_000000_Exp_Azimuth_Yaw_L2.fits
    └─ X01_PLX_G01_0003_000000_ProcessedEventdataSource_Det1_L2.lc
    └─ X01_PLX_G01_0003_000000_ProcessedEventdataSource_Det1_L2.pha
    └─ X01_PLX_G01_0003_000000_ProcessedEventdataSource_Det2_L2.lc
    └─ X01_PLX_G01_0003_000000_ProcessedEventdataSource_Det2_L2.pha
    └─ X01_PLX_G01_0003_000000_ProcessedEventdataSource_Det3_L2.lc
    └─ X01_PLX_G01_0003_000000_ProcessedEventdataSource_Det3_L2.pha
    └─ X01_PLX_G01_0003_000000_ProcessedEventdataSource_Det4_L2.lc
    └─ X01_PLX_G01_0003_000000_ProcessedEventdataSource_Det4_L2.pha
    └─ X01_PLX_G01_0003_000000_ProcessedEventdataSource_L2.lc

```

```

├── X01_PLX_G01_0003_000000_ProcessedEventDataSource_L2.pha
├── X01_PLX_G01_0003_000000_Src_Azimuth_Roll_L2.fits
├── X01_PLX_G01_0003_000000_Src_Azimuth_Yaw_L2.fits
└── X01_PLX_G01_0003_000000_WeightedRoll_L2.fits

```

1. **Light Curves and Spectra (.lc and .pha):** Light curves and spectra are generated from event mode data, light curves from count mode covering source, background, and Earth occultation intervals. For processed events, light curves and spectra are also produced separately for each individual detector.

Note that the count rate data is generated for a quick diagnosis of the overall count rate and variability. Simultaneous double events detected with POLIX (either between two different detectors, or between two channels of the same detector) are included in the count data while those are removed from the event data in Level1 processing. Therefore, the light curves generated from the count data and event data are not the same and should not be compared. The light curves generated from the count data are not meant for further scientific analysis. Please also note that POLIX is not a spectroscopic instrument and ARF and RMF are not currently generated for POLIX. Due to this, and for the fact that the reasons for background variation of POLIX are not yet fully understood, currently released POLIX data is not meant for spectroscopic analysis.

2. **Exp_Azimuth_Roll/Yaw_L2.fits:** FITS file containing azimuthal distribution of exposures. Azimuthal distribution of exposures for both Roll and Yaw-axis are calculated, but only Roll azimuthal distribution of exposures is used for further analysis in the pipeline.
3. **Src_Azimuth_Roll/Yaw_L2.fits:** FITS file containing azimuthal distribution of total counts for each one of the 48 anode cells. These products are not used in further analysis in the pipeline.
4. **EnergyRes_Src_Azimuth_Roll_L2.fits:** FITS file containing the counts of each azimuthal distribution and PHA (8192 values) for each of the 48 anode cells. This is a 3D FITS file or Data cube.

5. **WeightedRoll_L2.fits:** FITS file containing final intensity modulation with azimuthal angle obtained from POLIX. This is the final product required for polarization measurement. However, it includes azimuthal modulation of both source and background count rates.

Note: In Level2 data, parameter values in input text files in InputTxt folder depend on the options /selection given by the user while running the pipeline.

5 Installation and Setup

5.1. System Requirements

- **Operating System:** Linux
- **Architecture:** x86_64 (64-bit Linux)
- **Disk Space:** ≥ 12 GB free
- **RAM:** ≥ 4 GB (8 GB recommended)
- **Shell:** bash

5.2. Dependencies : Core Software

- **HEASoft** (version ≥ 6.32)
 - Includes FTOOLS and HEADAS libraries
 - Download: <https://heasarc.gsfc.nasa.gov/lheasoft/>
- **CFITSIO** (bundled with HEASoft or system-wide installation)

5.3. Python Environment

- **Python** (version ≥ 3.8)
- **Required packages:**
numpy scipy astropy matplotlib

To install the python packages use the following code in the terminal:

```
pip install numpy scipy astropy matplotlib
```

5.4 Download and Installation

Steps to install and set up the POLIX Level2 data processing pipeline on your system.

Step 1 — Download and locate the software

1. Download the package POLIXL2PIPELINE.V1.1.tar.xz from the provided source.
2. untar as below
tar -xvf POLIXL2PIPELINE.V1.1.tar.xz
3. Place the downloaded untar folder in a location of your choice (e.g., /home/user/).

Step 2 — Compile the source code

1. Open a terminal where the software is placed (e.g., /home/user/). Execute subsequent commands in the same terminal.
2. Navigate to the **polixpipeline/source** directory inside the package:
cd POLIXL2PIPELINE.V1.1/polixpipeline/source
3. Delete any pre-existing executables files and clean the build by running the command:
make clear
4. The POLIX Level2 pipeline comes with its own cfitsio library. The user has to add the location of the cfitsio folder to the LD_LIBRARY_PATH variable by the command given below. In our example the address to the POLIX cfitsio library folder will be “/home/user/POLIXL2PIPELINE.V1.1/polixpipeline/cfitsio”.
export LD_LIBRARY_PATH=<path to POLIX cfitsio folder>:\$LD_LIBRARY_PATH
5. Compile the source code to generate executable files in the bin folder inside polixpipeline. To compile use the below command from source directory:
make
6. Make sure the compilation occurs with no error.

Step 3 — Update Environment Variables

1. To access the pipeline commands from any location, update your bash file.
2. The address to the POLIX cfitsio library has to be added to the LD_LIBRARY_PATH variable. While the address to the POLIX software and the **bin** folder of the POLIX software has to be added to PATH. The POLIX pipeline itself can be sourced via the alias **POLIXL2PIPELINEV1P1**. The users are not to change the name of the variable **POLIXL2AZSOFTWARE**. Added below are the lines to be added to the **bashrc** file based on the aforementioned example:

```
export
```

```
LD_LIBRARY_PATH=/home/user/POLIXL2PIPELINE.V1.1/polixpipeline/cfitsio:$LD_LIBRARY_PATH
```

```
export POLIXL2AZSOFTWARE="/home/user/POLIXL2PIPELINE.V1.1/polixpipeline"
```

```
export PATH="$PATH:$POLIXL2AZSOFTWARE/bin"
```

```
alias POLIXL2PIPELINEV1P1=". $POLIXL2AZSOFTWARE/bin/polixl2pipeline-v1.1.sh"
```

3. Source the updated bash file using the command

```
source ~/.bashrc
```

Step 4 — Verify Installation

1. Run the pipeline using the alias:

POLIXL2PIPELINEV1P1

5.5 Pipeline Demonstration

- Copy the L1 dataset (e.g., X01_PLX_C24_0023_000000_L1_V1p1) to an empty directory (e.g., C24_0023_L2IN_Blanksy_06). Ensure the directory name includes the proposal ID and source name as suggested by the POC. Please refer to the user manual for the naming convention.
- Create an empty directory to save the L2 Out products (e.g., C24_0023_L2OUT_Blanksy_06). Ensure the directory name includes the proposal ID and source name as suggested by the POC.

POLIXPIPELINEL2 STARTS

```
=====
Running POLIX level2 pipeline
Task: POLIXPIPELINE Version:1P1 Release Date:04/08/2025
=====

Enter the Input Directory name ( Please give the full directory address where L1
data saved )
>/home/user/TEST/C24_0023_L2IN_Blanksy_06
Directory exists '/home/user/TEST/C24_0023_L2IN_Blanksy_06'

Enter the Output Directory name ( Please give the full directory address where
L2 data to be saved. Make sure this is an empty directory. )
>/home/user/TEST/C24_0023_L2OUT_Blanksy_06
Directory exists '/home/user/TEST/C24_0023_L2OUT_Blanksy_06'

Is this Data being processed for the first time?If yes pls type
'Y/y' or To proceed, please choose the L2out directory where
the Data Cube was last created , If you are processing for the
first time, simply enter 'Y/y'.
```

(As part of the pipeline, a data cube is generated. To avoid regenerating it in subsequent pipeline runs, you can provide the address of the previous L2 Out directory address where L2 products are saved (For our example case it will be

```
:/home/user/TEST/C24_0023_L2OUT_Blanksky_06)
```

```
>y
```

```
=====
The user confirmed with 'yes', indicating first time processing
of the data.
=====
```

```
User wants to give a gti file of your choice?
```

(If the user wants to give a GTI file of their own choice, they have to give the full directory address eg:/home/user/user.gti . “user.gti” should be fits file)

```
If yes give absolute directory address with filename or pls type
NONE
```

```
>none
```

```
user selected NONE option.
=====
=====
```

whether shift to be applied on a data cube ? *(As per POC suggestion give N/n ,for details please refer user manual Section_____)*

```
(Energy resolved azimuthal counts)
```

```
If yes please type Y/y or N/n
```

```
>n
```

```
=====
user selected None option.
```

User wants to give Det/Channel/PHA of your choice *(If user wants to give Det/Channel/PHA of their choice they can give here the full address of the Det/Channel/PHA selection file. In L1 directory*

eg:“/home/user/C24_0023_L2IN_Blanksky_06/X01_PLX_C24_0023_000000_L1_VIp1/Input Txt” we have file name called “sampleuserdetsidepha.txt” user can make copy of that and they can edit that text file)

```
If yes give absolute directory address with filename or pls type
no
```

```
>no
```

```
user selected NONE option.
=====
=====
```

POLIX level2 pipeline Parameters

L2 Input Directory

name:/home/user/TEST/C24_0023_L2IN_Blanksky_06

L2 Output Directory

name:/home/user/TEST/C24_0023_L2OUT_Blanksky_06

Data cube creation/selection:y

User Gti File:none

user selected option for shifting:n

User selected Det/Channel/PHA File:no

(User can confirm the inputs given above are correct before proceeding with the run of the PIPELINE)

=====

user wants to continue [y/n] : y

6 POLIX Level2 Software

The purpose of the Level2 software is to process the event based data to create a modulation curve of intensity as a function of the azimuthal angle of the Roll-axis with various user defined selection criteria . All the modules of Level2 software can be run in a pipeline.

Note: There are Input text files for each module.

6.1 polix_l2_lcpha:

Light curve extraction tool for event mode data and count mode data. Spectrum extraction tool for event mode data.

Input: Count mode data, event mode data, processed event, Time

Functional Description: This module contains two tools, polix-lc-count-creation.v1.0 to create light curves from the bin counter mode data and polix-lcpha-creation.v1.0 to create light curve and spectrum from the event mode data with various options for selection of detectors, channel, pulse height range, time window etc. Total no of counts for each bin satisfying all the selection criteria will be recorded for successive time bins that are within the valid time intervals for each type. For processed event data the pipeline creates, light curve and spectrum for each detector separately.

Note: There are Input text files for all lc and pha creation tools with various selection criteria. For description of Input text file please refer to Section 7.

Output: Light curve and spectrum fits files.

6.2 polix_l2_source_azimuth:

To create azimuthal distribution of counts for each anode cell of the four POLIX detectors.

Inputs:

- Source-azimuthal.txt: Contains the names of the input files along with detector channel and energy channel selection information. Located in the /InputTxt directory.
- X01_PLX_<proposals ID>_000000_ProcessedSourceUTEvents_L1.fits: Processed event file.

- **sshift_v4.txt:** Contains the energy-dependent azimuthal shifts applied to photons based on the anode in which they are detected. Located in the /caldb directory.

Functional Description: The POLIX processed event data contains information about the anode cell, total pulse height, and orientation of the satellite at the time of the X ray event. This information will be used to create the azimuthal distribution of total count with further selection of total pulse height, detector, channel, user GTI. There is an option for using the angular shift for each anode with the keyword “TRUE” or “FALSE” from the input text file.

Output:

X01_PLX_<Proposal ID>_000000_Src_Azimuth_Roll_L2.fits

X01_PLX_<Proposal ID>_000000_Src_Azimuth_yaw_L2.fits

FITS file containing azimuthal distribution of total counts for each one of the 48 anode cells.

6.3 polix_l2_exposure_azimuth:

To create azimuthal distribution of source exposure for the selected time window.

Input:

- Exposure-azimuthal.txt: Contains the names of the input files along with bin, GTI selection information. Located in the /InputTxt directory.
- X01_PLX_<proposal ID>_000000_AUX_L0_V1p1.att: Attitude file which contains satellite position information.
- X01_PLX_<proposal ID>_000000_FinalUT_gti_L1.gti: Time interval file

Functional Description: This module is to create an azimuthal distribution of source exposure during the source viewing operation. For the beginning and end of each bin of the azimuth angle, the pairs of entry times and exit times are determined from the satellite orbit file by interpolation. The difference between the exit time and the entry time gives the exposure for that azimuthal bin during one rotation of the satellite. Sum of all the exposures obtained during all the rotations of the satellite within the selected time window gives the total exposure. Same is done for all the azimuthal angle bins. This module also has an option of providing user GTI.

Output: FITS file containing azimuthal distribution of exposures.

6.4 polix_l2_EnergyRes_Shifted_Src_Azimuth:

To create energy resolved azimuthal distributions of source counts for each anode cell.

Input:

- **EnergyRes_Src_Azimuth.txt:** which contains the names of the input files along with detector channel and energy channel selection, user GTI information. Located in the /InputTxt directory.
- **X01_PLX_<proposals ID>_000000_ProcessedSourceUTEvents_L1.fits:** Processed event file.
- **sshift_v4.txt:** Contains the energy-dependent azimuthal shifts to be applied to photons based on the anode in which they are detected. Located in the /caldb directory.

Functional Description: The POLIX processed event data contains information about the anode cell, total pulse height, and orientation of the satellite at the time of the X ray event. This information will be used to create the azimuthal distribution of total count for each azimuthal angle for each anode cell with further selection on user GTI. It also has the option of applying appropriate shifts to the azimuthal count distribution of each anode cell so that, in the shifted distribution, the azimuthal angle represents the same quantity for all anodes: the angle of the detected photons relative to a fixed reference vector in sky coordinates. The information on whether the shift is applied or not is recorded in the header of the output FITS file and is later used to determine whether the modulation curves are already shifted or need shifting in step 6.5 (polix_l2_Weighted_Exposure).

Output:

X01_PLX_<proposals ID>_000000_EnergyRes_Src_Azimuth_Roll_L2.fits: FITS file containing the counts of each azimuthal distribution and PHA (8192 values) for each of the 48 anode cells for Roll-axis. This is a 3D FITS file or Data cube.

Note : Azimuthal angle varies from 0 to 360 degrees.

6.5 polix_l2_Weighted_Exposure

Generates the final azimuthal modulation curves (i.e., the azimuthal distribution of count rates) from all selected POLIX detector channels and energy channels.

Inputs:

- `l2_energyres_polarization_input.txt`:
Contains the names of the input files along with detector channel and energy channel selection information. Located in the /InputTxt directory.
- `X01_PLX_<propos ID>_000000_Exp_Azimuth_Roll_L2.fits`:
Output from the `polix_l2_Exposure_Azimuth` module.
- `X01_PLX_<propos ID>_000000_EnergyRes_Src_Azimuth_Roll_L2.fits`:
Output from the `polix_l2_EnergyRes_Shifted_Src_Azimuth` module. This is a 3D data array with dimensions $48 \times 8192 \times 360$.
- `sshift_v4.txt`:
Contains the energy-dependent azimuthal shifts applied to photons based on the anode in which they are detected. Located in the /caldb directory.

Functional Description: This module is written in python. For each selected anode and energy channel, the module calculates the azimuthal count-rate distribution - using the counts and exposure values from the input files - with respect to the Roll-axis angle from a fixed reference vector (azimuthal angle). It then applies the appropriate energy-dependent azimuthal shift (from `sshift_v4.txt`) based on the anode and photon energy. This ensures that the shifted azimuthal angle represents the angle of the detected photon from a fixed reference vector in the sky, independent of the satellite orientation. The count distributions are shifted only if indicated in the input header of the energy resolved azimuthal count distribution file (whether a shift has already been applied in Step 6.4 or not), while the exposure values are always shifted in this step. Finally, the module sums the shifted azimuthal count-rate distribution curves for all selected energies and anodes to produce the final azimuthal modulation curve.

Output:

`X01_PLX_<propos ID>_000000_WeightedRoll_L2.fits`: Final azimuthal modulation curve derived from all selected POLIX detector channels and selected energy channels.

POLIXL2HELP Command Overview

- To view the complete user manual: `polixl2help`
- To view help for a task: `polixl2help moduleName`

7. User Parameters

The POLIX Level2 pipeline uses multiple input user parameters. The user inputs can be provided via a text file or a FITS file, based on requirements.

Some default input parameters are kept in text files inside the InputTxt folder. It is advised that the files in this directory are not changed. The main input parameters that are provided to the pipeline are the channel selection and the user-defined GTIs. The InputTxt folder contains the following files:

det*lcpha.txt: These files contain the information of Detector 1 to 4 respectively used to make the lightcurve and spectrum files. The file contains the name of the input event file, the GTI file and the name of the output lightcurve and spectral files. The file also contains the time resolution for the light curve in seconds. The energy range of the light curve and spectrum is given in channel range. The **det.txt** file contains the information of the detectors to be used by the POLIX L2 pipeline.

lc--count.txt:** where “**” are “sc” for source, “bg” for background and “eo” for earth occultation, respectively. The files contain the name of the input event file, user and instrument GTI files, the name of the output lightcurve file and the detectors to use for the lightcurve generation.

Note: No PHA channel (energy) selection can be used for the light curve generation from count data.

lcpha--evnt.txt:** where “**” are “sc” for source, “bg” for background and “eo” for earth occultation, respectively. The files contain the details of the lightcurve and spectrum files that are to be created for the event files. The names of the input event file, the instrument and user GTI file, the name of the output lightcurve and spectrum files, the lightcurve bin size in seconds and the detectors to be used and their corresponding PHA (energy) range. The **lcpha-sc-Processed-evnt.txt** file contains similar information to the lightcurve and spectral file created for the source event file.

L2_energyres_polarization_input.txt: This file contains the input information for the polarisation modules. The file has the names of the output files of the intermittent products. The anode shift angle file name. It also contains the information of the detector and PHA range to use for the calculation of polarisation products.

sampleuserdetsidepha.txt: Is a sample file which contains the information of the information of lightcurve time bin size, as well as the information of the detector and PHA range to use for the calculation of polarisation products. A typical example is as follows:

```
1//lc_BinSize
0    0    0 8191//detector side pha range
0    1    0 8191
1    0    0 8191
1    1    0 8191
2    0    0 8191
2    1    0 8191
3    0    0 8191
3    1    0 8191
```

The PHA channel values go from 0 to 8191. The PHA value for 26 keV is 1800. The conversion from energy to PHA can be done linearly and is only valid between 8 and 30 keV.

Exposure-azimuthal.txt: Contains the input for the exposure calculation module. It contains the name of the input attitude file, instrument provided GTI and user provided GTI file names along with the output file names for the module and the azimuthal angle bin size.

EnergyRes_Src_Azimuth.txt: Similar to the **Source-azimuthal.txt** file, contains the name of the input file, GTI, shift file, the output file name, the size of azimuthal bin size, whether shift has been applied to the anodes, the detector channels, and PHA ranges to use.

Source-azimuthal.txt: Contains the name of the input file, GTI, shift file, the output azimuthal count files, the size of azimuthal bin size, whether shift has been applied to the anodes, the detector channels, and PHA ranges to use.

Note: DO NOT change the value of the azimuthal angle bin size in any of the input files.

8. Calibration Data & Associated Files

The POLIX Level2 pipeline requires three calibration files:

1. **Anode-Identifier.fits**

This file contains the charge ratio criteria used for identifying the anode on which the photon is incident in the POLIX detectors. The individual anodes in POLIX detectors are located at different positions in the azimuthal plane, and anode identification is crucial for determining the azimuthal position of the photon and obtaining the azimuthal modulation curve. This is applied in the Level1 pipeline.

2. **<obsid>_gain_para_normalized.txt**

This file contains the normalised gain values for all 48 anodes. The gains are normalised such that a PI channel number of 1800 always corresponds to around 26 keV. Photon energies can be then obtained using linear interpolation. This is applied in the Level1 pipeline.

3. **sshift_v4.txt**

This file contains the energy-dependent azimuthal shifts applied to photons based on the anode in which they are detected. The orbit file provides only the position of the Roll-axis in sky coordinates; we apply the appropriate shifts from sshift_v4.txt to each photon (based on its anode) so that the photon's position in sky coordinates - independent of the satellite's orientation - can be obtained.

These calibration files can be found in the **caldb** directory. They are crucial for obtaining the correct polarization products, and therefore it is advised not to change or move them.

8.1 Channel to energy conversion

The charge deposited at both ends of the anode wire in a POLIX detector is digitized by an ADC. The total charge deposited by an X-ray photon is proportional to the energy of the incident photon, and the sum of the digitized values from both ends corresponds to the respective energy channel. The energy channel values can range from 0 to 8191. Depending on the detector gain - which can be affected by factors such as temperature, gas pressure, etc - the energy channel

corresponding to a given photon energy can vary. POLIX is a self-calibrating instrument with respect to photon energy determination, as gain variations can be tracked using features in the background spectrum. Based on this monitoring, the energy channel values are normalized such that a normalized channel value of 1800 always corresponds to around 26 keV. Photon energies are then obtained from these normalized channel values using linear interpolation.

9. Barycenter correction

The POLIX instrument is in a low-Earth equatorial orbit, and the photon arrival times from a distant astrophysical source, as recorded by the detector, depend on the instrument's position and motion. Barycentric correction adjusts these arrival times to what they would be if measured at the solar system barycenter, effectively removing the effects of the satellite's motion around the Earth and the Earth's motion around the Sun. This is essential for precise timing studies.

While the POLIX data processing pipeline does not apply barycentric correction to the photon arrival times, the processed source event files generated by the Level1 pipeline, together with the spacecraft orbit file (.att), are compatible with the *barycorr* tool for applying barycenter corrections in HEASoft.

Example Code:

```
barycen infile=<input filename> outfile=<output filename> orbfile=<orbit filename>
orbext='ORBIT' orbform='COMPONENTS' orbcol='POSX,POSY,POSZ,VELX,VELY,VELZ'
refframe='FK5' timecol='TIME' ra=<source RA in deg> dec=<source DEC in deg>
startcol=START stopcol=STOP clobber=yes
```

10. Pipeline Walkthrough

10.1 Default Run of pipeline:

The following section shows the step-by-step run of the **POLIXL2PIPELINE** for one sample dataset. The data comprises multiple days of observation stretched over 4 sets of observation cycles. Before running the pipeline, make sure the latest version of the POLIX level2 software is properly installed and the full address of the POLIX Level1 directory is available. The user also has to create an empty directory where the final Level2 outputs will be placed after the pipeline has successfully completed the run. Make sure that the Level2 output directory is empty; otherwise, the pipeline will not run properly.

Once the **POLIXL2PIPELINE** is run, it will ask for the full directory address of the Level1 directory. Next, the user has to provide the full address of the Level2 pipeline. The pipeline will also ask if the pipeline is running for the first time. If the pipeline is run for the first time, then put “Y” as the input. If the pipeline is being rerun, some of the time-consuming products from the previous runs can be reused. In the case of subsequent runs, if the user wants to use some information from an older run, then the full address of the Level2 output directory needs to be given. Next, the user can choose to apply the energy-dependent shift to the anode counts at the data cube stage or later. The POC suggests this parameter to be “N”, i.e., the shift to be applied after the data cube generation. Next, the user also has the option to give a GTI file. The format of the GTI has to be properly implemented to use the GTI. The GTI file has to be a FITS file with two columns for start time (TSTART) and stop time (TSTOP). A typical example of the GTI is given in the Level1 folder as a *UTscience_gti_L1.gti file. The time in the TSTART and TSTOP columns is in seconds and is calculated as seconds elapsed from 1st January 2023 00:00:00.000 hours. The user can also provide a selection of detectors to use for the science products, like the light curve, spectrum and azimuthal count distribution for polarisation measurements. The user can also provide the time bin size of the same. A sample file is put in the Level1 folder in the name “sampleuserdetsidepha.txt”. For the default run, we do not use any user detector selection, so we put “no” here.

```

Enter the L2 Input Directory name
>/data1/plxusr1/Data/HER_L2IN/
Directory exists '/data1/plxusr1/Data/HER_L2IN/'

Enter the L2 Output Directory name
>/data1/plxusr1/Data/HER_L2OUT/
Directory exists '/data1/plxusr1/Data/HER_L2OUT/'

Is this Data being processed for first time?
If yes pls type 'Y/y' or To proceed, please choose the L2out directory where the Data Cube was last created
>Y
=====
User confirmed with 'yes',Indicating first time processing of the data.
=====

User wants to give gti file of your choice?

If yes give absolute directory adress with filename or pls type NONE
>NONE
user selcted NONE option.
=====

whether shift to be applied on data cube ?
(Energy resolved azimuthal counts)
If yes please type Y/y or N/n
>N
=====
user selcted None option.

User wants to give Det/Channel/PHA of your choice
If yes give absolute directory adress with filename or pls type no
>no
user selcted NONE option.

```

Figure 5: The inputs for the default run of the POLIX pipeline.

The data from the Level1 folder is then copied to the Level2 folder to run the individual Level2 modules. The user is then asked to confirm the information given for the running of the POLIX pipeline. The pipeline is terminated if the user inputs “n” and goes ahead if the user puts in “y”. A sample of the pipeline parameters check is given in Figure 6.

```

=====
POLIX level2 pipeline Parameters
L2 Input Directory name:/data1/plxusr1/Data/HER_L2IN/
L2 Output Directory name:/data1/plxusr1/Data/HER_L2OUT/
Data cube creation/selection:Y
User Gti File:NONE
user selected option for shifting:N
User selected Det/Channel/PHA File:no

=====
user wants to continue [y/n] : y
user has not choosen detector,side,pha value of its own
=====

```

Figure 6: Pipeline parameter confirmation

The pipeline next runs the individual modules based on the information given by the users; here, the default parameters are used. Once the polarisation module finishes running, the pipeline has also finished its run, and the following message is shown.

```
=====
[08/15/2025_02:38:33 PM] polix_l2_polarization module ends
=====*****=====
```

Figure 7: The last message that is shown at the end of the pipeline run.

After the pipeline has run successfully the folder structure as given in Section 4.3 will be created. The lightcurve is by default created with 1 second resolution in PHA channel 0 - 8191 i.e. all available channels. The spectrum is also created in the same PHA range.

Lightcurve: The lightcurve of the source obtained from the pipeline is plotted below in Figure 8 with a bin size of 100 seconds.

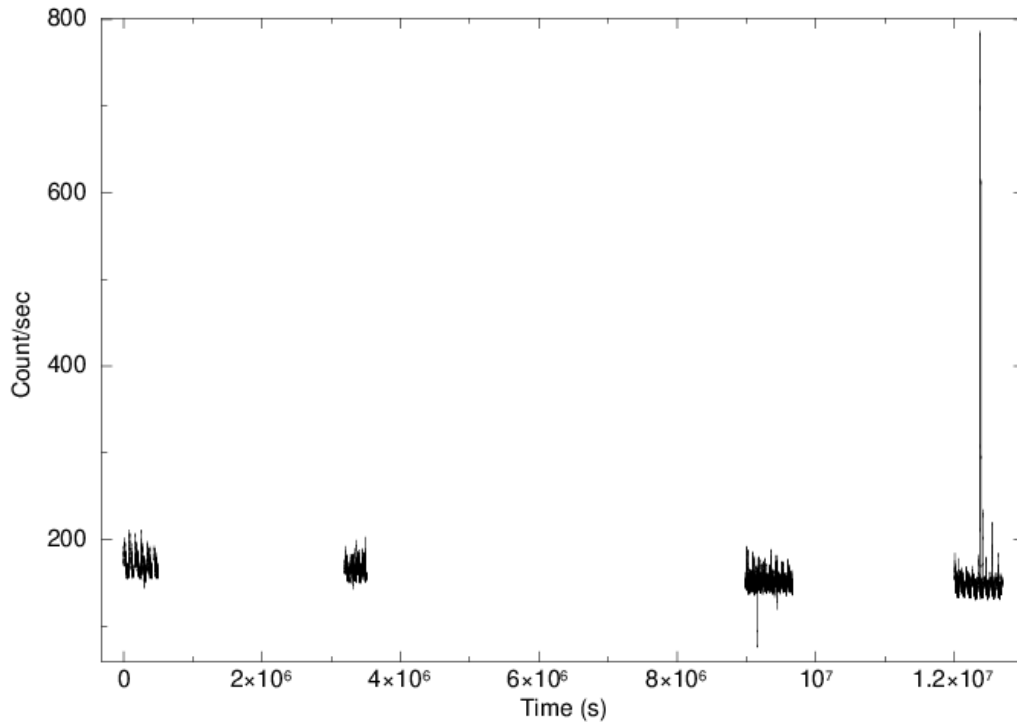


Figure 8: Lightcurve of the source covering time with 100 second bin size for Proposal ID: G01_0003.

Spectrum: The spectral information obtained from the source covering region is plotted in Figure 9 is shown in the PHA range of 0 - 8191. The calibration peak near channel 1800 is clearly seen in the image.

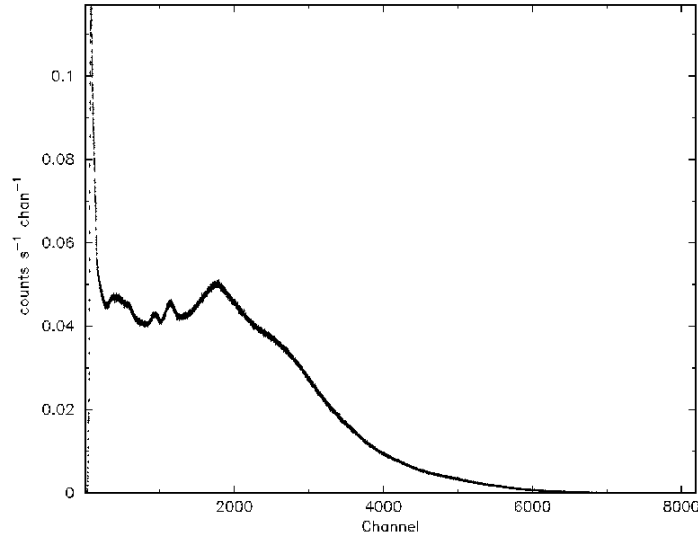


Figure 9: Spectrum of the source observation duration for Proposal ID: G01_0003.

Azimuthal distribution: The main output for extracting the polarization information of the source is the azimuthal distribution of count rate of the source. The azimuthal angular distribution of the count rate is shown in Figure 10. Many background effects are observed in the azimuthal distribution of count rate.

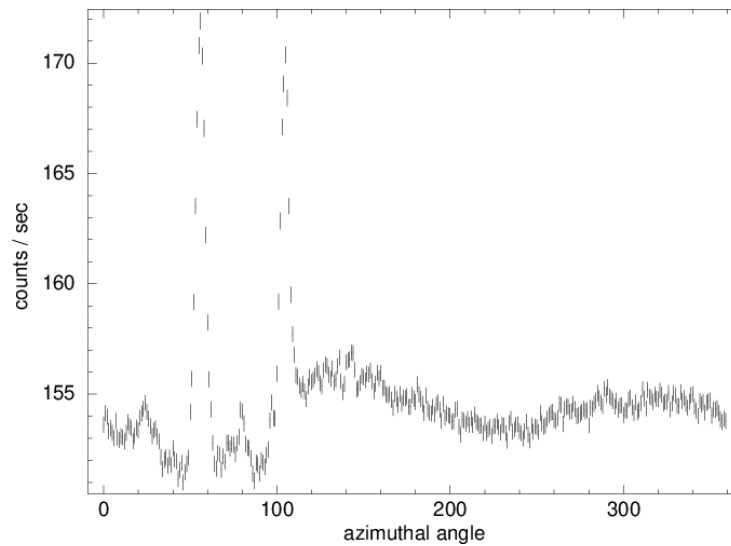


Figure 10: The azimuthal distribution of the count rate for Proposal ID: G01_0003.

10.2 Pipeline run with user inputs

The next walk-through of the pipeline contains the step-by-step guide to use an input GTI and detector selection to extract the output from the pipeline. We use a GTI with the filename “user_gti.gti” and the “userdetsidepha.txt” as the detector and PHA channel range.

```

Enter the L2 Input Directory name
>/data1/plxusr1/Data/HER_L2IN/
Directory exists '/data1/plxusr1/Data/HER_L2IN/'

Enter the L2 Output Directory name
>/data1/plxusr1/Data/HER_L2OUT_custom_run/
Directory exists '/data1/plxusr1/Data/HER_L2OUT_custom_run/'

Is this Data being processed for first time?
If yes pls type 'Y/y' or To proceed, please choose the L2out directory where the Data Cube was last created
>Y
=====
User confirmed with 'yes',Indicating first time processing of the data.
=====

User wants to give gti file of your choice?

If yes give absolute directory address with filename or pls type NONE
>/data1/plxusr1/Data/user_gti.gti
File exists '/data1/plxusr1/Data/user_gti.gti'
user selected gti of own choice.
=====

whether shift to be applied on data cube ?
(Energy resolved azimuthal counts)
If yes please type Y/y or N/n
>N
=====
user selcted None option.

User wants to give Det/Channel/PHA of your choice
If yes give absolute directory address with filename or pls type no
>/data1/plxusr1/Data/userdetsidepha.txt
File exists '/data1/plxusr1/Data/userdetsidepha.txt'
user selected channel/PHA of own choice.

```

Figure 11: The inputs used for a custom run of the POLIX pipeline

The user can also check whether the inputs are correctly recorded to make sure the pipeline is going to perform in the modified conditions the user wants.

```

=====
POLIX level2 pipeline Parameters
=====
L2 Input Directory name:/data1/plxusr1/Data/HER_L2IN/
L2 Output Directory name:/data1/plxusr1/Data/HER_L2OUT_custom_run/
Data cube creation/selection:Y
User Gti File:/data1/plxusr1/Data/user_gti.gti
user selected option for shifting:N
User selected Det/Channel/PHA File:/data1/plxusr1/Data/userdetsidepha.txt
=====

```

Figure 12 : Custom selections are shown while running the pipeline with multiple user input files.

Structure of User Input Files: In order to run the pipeline with custom conditions, the user needs to provide additional input files in specific formats. These files allow the pipeline to restrict the analysis to certain time intervals, detectors, and energy channel ranges, and also define parameters such as the light curve bin size.

user_gti.gti

The file `user_gti.gti` is used when the user wishes to apply custom Good Time Intervals (GTIs) for a specific analysis.

- **File type:** FITS
- **Relevant extension:** HDU[1] (the GTI extension)
- **Contents:** Two columns, `START` and `STOP`, which define the beginning and end times of each GTI window.
- **Units:** Times are expressed in seconds, measured relative to the reference epoch `MJDREF = 1 January 2023`.
- **Usage:** The pipeline restricts data extraction and processing only to events lying within these GTIs.

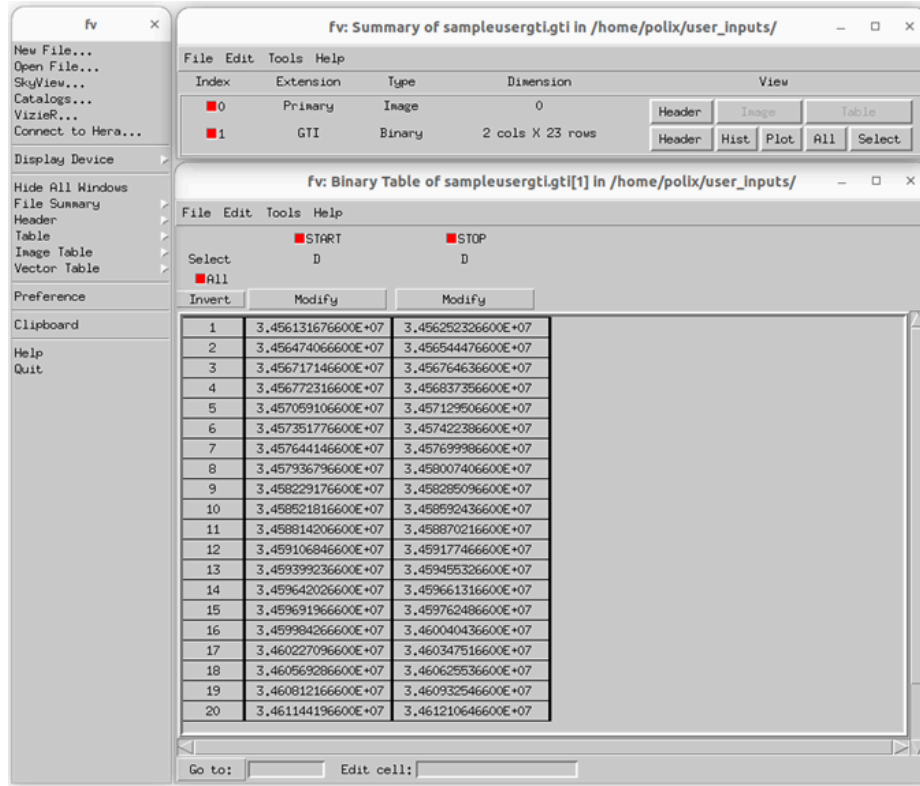


Figure 13: A screenshot of a sample user GTI file.

userdetsidepha.txt

The file userdetsidepha.txt is a plain-text configuration file that specifies:

- **First line:** Defines the light curve bin size in seconds, prefixed by the keyword //lc_BinSize.
- **Subsequent lines:** Each row corresponds to one detector-channel combination and its associated PHA range.
- **Format of detector-channel-PHA lines:** <detector_ID> <channel_ID> <PHA_min>
<PHA_max>

detector_ID: Detector number (0-3)

channel_ID: Channel number (0 or 1)

PHA_min, PHA_max: Lower and upper bounds of the PHA channel selection
(allowed range: 0-8191)

```
polix@polixppcg:~/user_inputs$ more sampleuserdetsidepha.txt
1//lc_BinSize
0      0      480      1800//dectector side pha range
0      1      480      1800
1      0      480      1800
1      1      480      1800
2      0      480      1800
2      1      480      1800
3      0      480      1800
3      1      480      1800
```

Figure 14 :A sample userdetsidepha.txt file.

In this example

- The light curve bin size is **1 second**.
- All four detectors (0-3) and both channels (0,1) are included.
- The selected PHA range is **480–1800** for all detector-side combinations.

Only the lines corresponding to the detector–channel combinations that need to be processed should be kept in the file. For example, if only detector 2 (both channels) is to be processed, then only the lines.

```
1      0      480      1800
1      1      480      1800
```

need to be present.

10.3 Running individual POLIX Modules:

In this section, we give details of the individual modules of the POLIX Level2 PIPELINE. The individual modules can be used to extract lightcurve, spectrum or other polarisation products with customised/user-provided input parameters. To successfully run the individual modules, the relevant input text files, which contain the details of the files required for running the module and the files which are mentioned in the input text file must be in the same folder. The output file would also be placed in the same folder.

10.3.1. Lightcurve and spectrum module:

The lightcurve and spectrum module makes the lightcurve and spectrum files. It requires a lcpha-**-evnt.txt along with input files in the same folder. The inputs are: event file, GTI files and if the user wants a user defined GTI. Time bin size and PHA range can be provided via the input text file.

The code is run as :

polix-lcpha-creation.v1.0 < parameter_filename.txt (event data)

polix-lc-count-creation.v1.0 < parameter_filename.txt (count data)

An example output for running the polix-lc-count-creation.v1.0 module has been shown in Figure 15 and for the polix-lcpha-creation.v1.0 module is shown in Figure 16.

```

No of entries      8
  0                0
  0                1
  1                0
  1                1
  2                0
  2                1
  3                0
  3                1
  65
Total no of events processed: 39052523
38539470.070376359          51231324.793093048

```

Figure 15: An example output for the count lightcurve creation module.

```

No of entries      8
      0      0      0      8191
      0      1      0      8191
      1      0      0      8191
      1      1      0      8191
      2      0      0      8191
      2      1      0      8191
      3      0      0      8191
      3      1      0      8191
      0
      67
Total no of events processed: 81213471

```

Figure 16: An example output for the event lightcurve and spectrum creation module.

Detailed explanation:

Here, we present a detailed walk-through of the POLIX polix-lc-count-creation.v1.0 module, which is used for creation of the lightcurve and spectrum file from a given event file. Here we will run the module on the event file, without any external GTI, for all POLIX detectors and all channels available across the entire PHA range. To run the module on the event data, we use the input text file “lcpha-sc-evnt.txt“. In Figure 17, we see the contents of the input file. To run the module, we therefore require:

1. The event file for the source covering duration
“X01_PLX_G01_0003_000000_SourceUTevents_L1.fits”,
2. The instrument GTI file “X01_PLX_G01_0003_000000_FinalUT_gti_L1.gti”,
3. Any user-defined GTIs, in this case “none”,
4. The names of the output lightcurve
“X01_PLX_G01_0003_000000_EventdataSource_L2.lc”, and
5. The spectrum file “X01_PLX_G01_0003_000000_EventdataSource_L2.pha”

Below these, we have the time resolution of the lightcurve and the extension of the GTI file, which has to be used; in this case, it is the second extension. Next, we have detector, channel and PHA range selection, which is written as <detector number> (0-3), <channel number> (0-1), PHA minimum value and PHA maximum value (0-8191). In this case, the input files, namely the “X01_PLX_G01_0003_000000_SourceUTevents_L1.fits”,

“X01_PLX_G01_0003_000000_FinalUT_gti_L1.gti”, and “lcpha-sc-evnt.txt” file must be in the directory where the pipeline module is to be run.

Next, the module can be executed using the command:

polix-lcpha-creation.v1.0 < lcpha-sc-evnt.txt

The output for this run is given in Figure 15.

In summary, identify a template text file format for the POLIX module, copy the relevant files and the text file to a directory and run the module in the same directory as mentioned in Section 10.3. The output files would also be placed in the same directory.

```
X01_PLX_G01_0003_000000_SourceUTevents_L1.fits //Input file
X01_PLX_G01_0003_000000_FinalUT_gti_L1.gti // Input gti file
NONE //user gti file
X01_PLX_G01_0003_000000_EventdataSource_L2.lc //output lc filename1
X01_PLX_G01_0003_000000_EventdataSource_L2.pha //output pha filename2
1 //lc BinSize
2 //Extension number in gti file
0 0 0 8191//detector side pha range
0 1 0 8191
1 0 0 8191
1 1 0 8191
2 0 0 8191
2 1 0 8191
3 0 0 8191
3 1 0 8191
```

Figure 17: Content of the input text file for polix-lc-count-creation.v1.0 module

10.3.2. Source azimuth module:

This module is to create an azimuthal distribution of total counts for each of the 12 anode cells for the four POLIX detectors, i.e. for a total of 48 anode cells. This module will also allow selection of total pulse height, detector, channel and time interval. The example test file Source-azimuthal.txt can be used as reference. The input event file, GTI files, the energy-dependent shift for each anode and if the user wants a user defined GTI. The PHA range can be provided via the input text file. The output files can also be given as input parameters.

The code is run as:

polix_l2_srce_azi.v1.1 < parameter_filename.txt

An example output for running the polix_l2_srce_azi.v1.1 module has been shown in Figure 18.

```

updated code
      0      0      0      8191
      0      1      0      8191
      1      0      0      8191
      1      1      0      8191
      2      0      0      8191
      2      1      0      8191
      3      0      0      8191
      3      1      0      8191
No of entries      8
RA OBJ,DEC OBJ
 254.45800000000000      35.341999999999999
yaw azimuthal
stime,etime 38539469.070376001      51231326.230999999

```

Figure 18: An example output for the source azimuth module.

10.3.3. Exposure module:

The exposure module calculates the exposure in each angular bin size of 1 degree. The example test file Exposure-azimuthal.txt can be used as reference. The inputs are: attitude file, GTI files and if the user wants a user defined GTI. Time bin size and PHA range can be provided via the input text file. The output files can also be given as input parameters.

The code is run as:

polix_l2_exposure_azimuth.v1.1< parameter_filename.txt

An example output for running the polix_l2_exposure_azimuth.v1.1 module has been shown in Figure 19.

```

updated code
no of source entries      626
final Gti
No of entries in GTI file:      626
source Ra and Dec values
254.4580000000000000      35.3419999999999999
No of entries in attitude file: 21147683
No of entries in Det 1 GTI file: 1952
No of entries in Det 2 GTI file: 1952
No of entries in Det 3 GTI file: 1952
No of entries in Det 4 GTI file: 1952
merge source GTI and Det1 GTI and then      calculate roll and Yaw exposure for Det1
Total exposure: 527454.438
merge source GTI and Det2 GTI and then      calculate Roll and Yaw exposure for Det2
Total exposure: 527454.438
merge source GTI and Det3 GTI and then      calculate Roll and Yaw exposure for Det3
Total exposure: 527454.438
merge source GTI and Det4 GTI and then      calculate Roll and Yaw exposure for Det4
Total exposure: 527454.438

```

Figure 19: An example output for the exposure module.

10.3.4. Energy dependent source azimuth module:

This module is to create an azimuthal distribution of total counts for each of the 12 anode cells for the four POLIX detectors, i.e. for a total of 48 anode cells but the shifts are now energy dependent. The code is run as:

polix_l2_EnergyRes_Shifted_Src_Azimuth_v1.1 < EnergyRes_Src_Azimuth.txt

An example output for running the polix_l2_EnergyRes_Shifted_Src_Azimuth_v1.1 module has been shown in Figure 20.

```

updated code
      0      0      0      8191
      0      1      0      8191
      1      0      0      8191
      1      1      0      8191
      2      0      0      8191
      2      1      0      8191
      3      0      0      8191
      3      1      0      8191
No of entries      8
RA_OBJ,DEC_OBJ
254.4580000000000000      35.3419999999999999

```

Figure 20: An example output for the energy dependent source azimuth module.

10.3.5. Weighted exposure module

The weighted module calculates the count rate of the source as a function of azimuthal angle with an angular bin size of 1 degree. The example input parameter text file `l2_energyres_polarization_input.txt` can be used as a reference. The exposure file as well as the source count files as a function of azimuthal angles, and the shift file name are taken as inputs. The PHA range can also be provided via the input text file.

The code is run as:

`l2_polarization_v3 < parameter_filename.txt`

11. Timeline of POLIX detectors

After a few months of in-flight operation, from 25th May, 2024, the Channel-2 of Detector-4 showed an increase in resistance at one end of one channel. As a result, identification of anodes by the process of charge division is not possible for this channel any more. Therefore, data from Channel-2 of Detector-4 is not usable after 25th May, 2024.

On 14th November 2024, Detector-3 developed an anomaly. The anti-coincidence counts are reduced, reducing the background rejection efficiency of this detector. As a result, there is a large increase in the background count rate in Detector-3 and data from this detector is not usable from 14th November 2024.

The gas gains of three of the detectors (1, 2 and 3) have remained stable since launch while Detector-4 showed a gradual increase in the detector gain, indicating a slow loss of detector gas at the rate of about 15% per year. The gas gain is adjusted periodically by reducing the high voltage in this detector.

12. Variability of background modulation and limitations

A large number of blank sky regions have been observed with POLIX for extended periods. The azimuthal distribution of the background is found to have a small two-fold modulation, with the modulation fraction varying between 0.5-1.5%. Azimuthal distribution of the background counts has been examined for its possible dependence on various parameters like declination, sun-angle, azimuthal distribution of the anti-coincidence count rate etc. However, the amplitude and phase of the background modulation is variable and it is not found to depend on any particular parameter.

In addition, for observations in a few directions in the sky, narrow spikes are observed in certain azimuthal angle ranges (See Fig 10). For a given source observation, the spikes appear at the same orientation of the detector wires with respect to the source direction. For observations that are carried out at different times of the year, and at different locations of the spacecraft over the earth, the spike characteristics are the same.

In the presence of this variable background modulation, and that of the spikes in some observations, X-ray polarization measurement is not possible with the currently released data, in which a source is observed for a few days and multiple blank sky regions are observed for additional days. A new strategy of observing a bright X-ray source and nearby blank sky regions has been adopted for polarization measurements with POLIX. Data from these observations and methods for background subtraction will be detailed in future data/handbook release.

The POLIX measurements of background modulations can be useful to constrain models of local charged particle anisotropy with respect to the local magnetic field configuration.

Several Gamma-ray bursts have been detected with POLIX. Timing characteristics of these can be studied, and source localisation can be attempted along with the data from other GRB detectors.

A Annexure

A1 Fits File description

A1.1 L1 products: This section describes the contents of the FITS L1 products (Tables 4 - 9).

Table 4: Description of L1 data product: Event File

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: Events				
1	TIME	Float (64-bit)	S	UTC (s since POLIX epoch) of event occurrence
2	CHANNEL	Unsigned integer (8-bit)	ID(0-3)	Channel in which event occurred
3	DETECTOR	Unsigned integer (8-bit)	ch(0,1)	Detector in which event occurred
4	PHA1	Unsigned integer(16-bit)	-	Pulse Height Analyser 1
5	PHA2	Unsigned integer(16-bit)	-	Pulse Height Analyser 2
6	ANODE	Unsigned integer(16-bit)	-	Anode number
Extension 2: GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time

The structure and contents of the Event File are identical for source, background, and Earth occultation observations. Therefore, the description provided in this table is applicable to all three cases. The second extension in each file contains the GTI (Good Time Intervals) corresponding to the respective source, background, or Earth occultation observation.

Table 5: Description of L1 data product: Count File

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
1	TIME	Float (64-bit)	S	UTC (s since POLIX epoch) of event occurrence
2	CHANNEL	Unsigned integer (8-bit)	ID(0-3)	Channel in which event occurred
3	DETECTOR	Unsigned integer (8-bit)	ch(0,1)	Detector in which event occurred
4	ACNT	Unsigned integer (8-bit)		Accepted counts
5	LLD2	Unsigned integer (8-bit)		Lower Level Discriminator 2
6	LLD1	Unsigned integer (8-bit)		Lower Level Discriminator 1
7	ANTI	Unsigned integer (8-bit)		Anti-coincidence counts
8	ULD2	Unsigned integer (8-bit)		Upper Level Discriminator 2
9	ULD1	Unsigned integer (8-bit)		Upper Level Discriminator 1
Extension 2: GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time

The structure and contents of the Count File are identical for source, background, and Earth occultation observations. Therefore, the description provided in this table is applicable to all three cases. The second extension in each file contains the GTI (Good Time Intervals) corresponding to the respective source, background, or Earth occultation observation.

Table 6: Description of L1 data product: Final_GTI File

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: Source GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time
Extension 2: Background GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time
Extension 3: Earth Occultation GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time
Extension 4: Detector1 GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time
Extension 5: Detector2 GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time
Extension 6: Detector3 GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time
Extension 7: Detector4 GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time

Table 7: Description of L1 data product: Science_GTI File

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time

Table 8: Description of L1 data product: Housekeeping File

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: polix_hk_data				
1	TIME	Float (64-bit)	S	UTC (s since POLIX epoch) of event occurrence
2	FE1_HV_MON	Float (32-bit)	-	HV value of Detector 01
3	FE2_HV_MON	Float (32-bit)	-	
4	FE3_HV_MON	Float (32-bit)	-	
5	FE4_HV_MON	Float (32-bit)	-	
6	DET1_C1_LLD1_CTR	Float (32-bit)	-	Counts below LLD threshold in Detector 01 Channel 01
7	DET1_C1_LLD2_CTR	Float (32-bit)	-	
8	DET1_C2_LLD1_CTR	Float (32-bit)	-	
9	DET1_C2_LLD2_CTR	Float (32-bit)	-	
10	DET2_C1_LLD1_CTR	Float (32-bit)	-	
11	DET2_C1_LLD2_CTR	Float (32-bit)	-	
12	DET2_C2_LLD1_CTR	Float (32-bit)	-	
13	DET2_C2_LLD2_CTR	Float (32-bit)	-	
14	DET3_C1_LLD1_CTR	Float (32-bit)	-	
15	DET3_C1_LLD2_CTR	Float (32-bit)	-	
16	DET3_C2_LLD1_CTR	Float (32-bit)	-	
17	DET3_C2_LLD2_CTR	Float (32-bit)	-	
18	DET4_C1_LLD1_CTR	Float (32-bit)	-	
19	DET4_C1_LLD2_CTR	Float (32-bit)	-	
20	DET4_C2_LLD1_CTR	Float (32-bit)	-	
21	DET4_C2_LLD2_CTR	Float (32-bit)	-	
22	FE1_HV_REF_MON	Float (32-bit)	-	Reference HV values of Detector 01
23	FE2_HV_REF_MON	Float (32-bit)	-	
24	FE3_HV_REF_MON	Float (32-bit)	-	
25	FE4_HV_REF_MON	Float (32-bit)	-	
26	DET1_C1_ULD1_CTR	Float (32-bit)	-	Counts above ULD threshold in Detector 01 Channel 01
27	DET1_C1_ULD2_CTR	Float (32-bit)	-	
28	DET1_C2_ULD1_CTR	Float (32-bit)	-	
29	DET1_C2_ULD2_CTR	Float (32-bit)	-	
30	DET2_C1_ULD1_CTR	Float (32-bit)	-	

31	DET2_C1_ULD2_CTR	Float (32-bit)	-	
32	DET2_C2_ULD1_CTR	Float (32-bit)	-	
33	DET2_C2_ULD2_CTR	Float (32-bit)	-	
34	DET3_C1_ULD1_CTR	Float (32-bit)	-	
35	DET3_C1_ULD2_CTR	Float (32-bit)	-	
36	DET3_C2_ULD1_CTR	Float (32-bit)	-	
37	DET3_C2_ULD2_CTR	Float (32-bit)	-	
38	DET4_C1_ULD1_CTR	Float (32-bit)	-	
39	DET4_C1_ULD2_CTR	Float (32-bit)	-	
40	DET4_C2_ULD1_CTR	Float (32-bit)	-	
41	DET4_C2_ULD2_CTR	Float (32-bit)	-	
42	DET1_C1_ANTI_CTR	Float (32-bit)	-	Counts recorded by the ANTI-coincidence layer in Detector 01
43	DET1_C1_ACC_CTR	Float (32-bit)	-	Accepted Counts in Detector 01
44	DET1_C2_ANTI_CTR	Float (32-bit)	-	
45	DET1_C2_ACC_CTR	Float (32-bit)	-	
46	DET2_C1_ANTI_CTR	Float (32-bit)	-	
47	DET2_C1_ACC_CTR	Float (32-bit)	-	
48	DET2_C2_ANTI_CTR	Float (32-bit)	-	
49	DET2_C2_ACC_CTR	Float (32-bit)	-	
50	DET3_C1_ANTI_CTR	Float (32-bit)	-	
51	DET3_C1_ACC_CTR	Float (32-bit)	-	
52	DET3_C2_ANTI_CTR	Float (32-bit)	-	
53	DET3_C2_ACC_CTR	Float (32-bit)	-	
54	DET4_C1_ANTI_CTR	Float (32-bit)	-	
55	DET4_C1_ACC_CTR	Float (32-bit)	-	
56	DET4_C2_ANTI_CTR	Float (32-bit)	-	
57	DET4_C2_ACC_CTR	Float (32-bit)	-	
58	FE1_HV_DCDC_12VP_MON	Float (32-bit)	-	HV DCDC +12V value for Detector 01
59	FE2_HV_DCDC_12VP_MON	Float (32-bit)	-	
60	FE3_HV_DCDC_12VP_MON	Float (32-bit)	-	
61	FE4_HV_DCDC_12VP_MON	Float (32-bit)	-	
62	FE1_CSPA_12VP_MON	Float (32-bit)	-	CSPA +12V value for Detector 01
63	FE2_CSPA_12VP_MON	Float (32-bit)	-	
64	FE3_CSPA_12VP_MON	Float (32-bit)	-	
65	FE4_CSPA_12VP_MON	Float (32-bit)	-	
66	BE1_DC_DC1_5VP_MON	Float (32-bit)	-	DCDC +5V value for BE 01
67	BE2_DC_DC1_5VP_MON	Float (32-bit)	-	
68	BE3_DC_DC1_5VP_MON	Float (32-bit)	-	
69	BE4_DC_DC1_5VP_MON	Float (32-bit)	-	
70	BE1_DC_DC1_15VP_MON	Float (32-bit)	-	DCDC +15V value for BE 01
71	BE2_DC_DC1_15VP_MON	Float (32-bit)	-	

72	BE3_DC_DC1_15VP_MON	Float (32-bit)	-	
73	BE4_DC_DC1_15VP_MON	Float (32-bit)	-	
74	BE1_DC_DC1_15VN_MON	Float (32-bit)	-	DCDC -15V value for BE 01
75	BE2_DC_DC1_15VN_MON	Float (32-bit)	-	
76	BE3_DC_DC1_15VN_MON	Float (32-bit)	-	
77	BE4_DC_DC1_15VN_MON	Float (32-bit)	-	
78	DET1_C1_LLD_MON	Float (32-bit)	-	LLD Threshold for Detector 01 Channel 01
79	DET1_C1_ULD_MON	Float (32-bit)	-	ULD Threshold for Detector 01 Channel 01
80	DET1_C2_LLD_MON	Float (32-bit)	-	
81	DET1_C2_ULD_MON	Float (32-bit)	-	
82	DET1_ANTI_LLD_MON	Float (32-bit)	-	ANTI-LLD Threshold for Detector 01
83	DET2_C1_LLD_MON	Float (32-bit)	-	
84	DET2_C1_ULD_MON	Float (32-bit)	-	
85	DET2_C2_LLD_MON	Float (32-bit)	-	
86	DET2_C2_ULD_MON	Float (32-bit)	-	
87	DET2_ANTI_LLD_MON	Float (32-bit)	-	
88	DET3_C1_LLD_MON	Float (32-bit)	-	
89	DET3_C1_ULD_MON	Float (32-bit)	-	
90	DET3_C2_LLD_MON	Float (32-bit)	-	
91	DET3_C2_ULD_MON	Float (32-bit)	-	
92	DET3_ANTI_LLD_MON	Float (32-bit)	-	
93	DET4_C1_LLD_MON	Float (32-bit)	-	
94	DET4_C1_ULD_MON	Float (32-bit)	-	
95	DET4_C2_LLD_MON	Float (32-bit)	-	
96	DET4_C2_ULD_MON	Float (32-bit)	-	
97	DET4_ANTI_LLD_MON	Float (32-bit)	-	
98	POLIX_BE_1_M_TEMP	Float (32-bit)	-	BE 01 Temperature (Main)
99	POLIX_BE_1_R_TEMP	Float (32-bit)	-	BE 01 Temperature (Redundant)
100	POLIX_BE_2_M_TEMP'	Float (32-bit)	-	
101	POLIX_BE_3_M_TEMP'	Float (32-bit)	-	
102	POLIX_BE_3_R_TEMP	Float (32-bit)	-	
103	POLIX_BE_4_M_TEMP	Float (32-bit)	-	
104	POLIX_BE_4_R_TEMP	Float (32-bit)	-	
105	POLIX_COLLEMATOR_1_TEMP	Float (32-bit)	-	Collimator Temperature 01
106	POLIX_COLLEMATOR_2_TEMP	Float (32-bit)	-	Collimator Temperature 02
107	SCATTERER_TEMP	Float (32-bit)	-	Scatterer Temperature
108	POLIX_DET_HSG1_ENSEDE_TO P_TEMP	Float (32-bit)	-	Detector 01 Inside Temperature (Top)
109	POLIX_DET_HSG1_ENSEDE_BO T_TEMP	Float (32-bit)	-	Detector 01 Inside Temperature (Bottom)
110	POLIX_DET_HSG1_OUT_TEMP	Float (32-bit)	-	Detector 01 Outside Temperature

111	POLIX_DET_HSG2_ENSEDE_TO_P_TEMP	Float (32-bit)	-	
112	POLIX_DET_HSG2_ENSEDE_BO_T_TEMP	Float (32-bit)	-	
113	POLIX_DET_HSG2_OUT_TEMP	Float (32-bit)	-	
114	POLIX_DET_HSG3_ENSEDE_TO_P_TEMP	Float (32-bit)	-	
115	POLIX_DET_HSG3_ENSEDE_BO_T_TEMP	Float (32-bit)	-	
116	POLIX_DET_HSG3_OUT_TEMP	Float (32-bit)	-	
117	POLIX_DET_HSG4_ENSEDE_TO_P_TEMP	Float (32-bit)	-	
118	POLIX_DET_HSG4_ENSEDE_BO_T_TEMP	Float (32-bit)	-	
119	POLIX_DET_HSG4_OUT_TEMP	Float (32-bit)	-	
120	FE3_HV_STAT	Unsigned integer (8-bit)	-	HV Status for Detector 03
121	FE2_CRN	Unsigned integer (8-bit)	-	Corona Status for Detector 02
122	FE4_HV_STAT	Unsigned integer (8-bit)	-	
123	FE2_HV_STAT	Unsigned integer (8-bit)	-	
124	FE1_CRN	Unsigned integer (8-bit)	-	
125	FE4_CRN	Unsigned integer (8-bit)	-	
126	SLEW_RATE_COMP	Float (32-bit)	-	Slew Rate
127	FE1_SAA	Unsigned integer (8-bit)	-	
128	FE3_CRN	Unsigned integer (8-bit)	-	
129	FE3_SAA	Unsigned integer (8-bit)	-	
130	FE4_SAA	Unsigned integer (8-bit)	-	
131	FE1_HV_STAT	Unsigned integer (8-bit)	-	
132	FE2_SAA	Unsigned integer (8-bit)	-	

Table 9: Description of L1 data product: Attitude file

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: polix_att_data				
1	TIME	Float (64-bit)	S	UTC (s since POLIX epoch) of event occurrence
2	POINTING_RA	Float (32-bit)	deg	Right ascension of source pointing direction
3	POINTING_DEC	Float (32-bit)	deg	Declination of source pointing direction
4	ROLL_RA	Float (32-bit)	deg	Right ascension of spacecraft Roll-axis
5	ROLL_DEC	Float (32-bit)	deg	Declination of spacecraft Roll-axis
6	PITCH_RA	Float (32-bit)	deg	Right ascension of spacecraft Pitch-axis

7	PITCH_DEC	Float (32-bit)	deg	Declination of spacecraft Pitch-axis
8	YAW_RA	Float (32-bit)	deg	Right ascension of spacecraft Yaw-axis
9	YAW_DEC	Float (32-bit)	deg	Declination of spacecraft Yaw-axis
10	POSX	Float (32-bit)	km	Spacecraft position in ECI frame (X coordinate)
11	POSY	Float (32-bit)	km	Spacecraft position in ECI frame (Y coordinate)
12	POSZ	Float (32-bit)	km	Spacecraft position in ECI frame (Z coordinate)
13	VELX	Float (32-bit)	km/s	X component of spacecraft velocity vector
14	VELY	Float (32-bit)	km/s	Y component of spacecraft velocity vector
15	VELZ	Float (32-bit)	km/s	Z component of spacecraft velocity vector
16	EARTH_LAT	Float (32-bit)	deg	Latitude of the spacecraft footprint on Earth
17	EARTH_LON	Float (32-bit)	deg	Longitude of the spacecraft footprint on Earth
18	ALTITUDE	Float (32-bit)	deg	Spacecraft altitude above the Earth's surface
19	VELOCITY_RA	Float (32-bit)	deg	Right ascension of the spacecraft velocity vector
20	VELOCITY_DEC	Float (32-bit)	deg	Declination of the spacecraft velocity vector
21	SUN_RA	Float (32-bit)	deg	Right ascension of Sun
22	SUN_DEC	Float (32-bit)	deg	Declination of Sun
23	MOON_RA	Float (32-bit)	deg	Right ascension of Moon
24	MOON_DEC	Float (32-bit)	deg	Declination of Moon
25	SUN_ANGLE	Float (32-bit)	deg	Angle between target and Sun
26	MOON_ANGLE	Float (32-bit)	deg	Angle between target and Moon
27	ELV	Float (32-bit)	deg	Elevation of target from Earth horizon
28	ANG_OFFSET	Float (64-bit)	-	Angle between Pitch-axis and pointing direction
29	SUN_EARTH_ANGLE	Float (64-bit)	-	Earth-centered Sun-Spacecraft angle
30	MOON_EARTH_ANGLE	Float (64-bit)	-	Earth-centered Moon-Spacecraft angle
31	QUAL_FLAG	Unsigned integer (8-bit)	-	Quality Flag

A1.2 L2 products: This section describes the contents of the FITS L2 products (Tables 10 - 17).

Table 10: Description of L2 data product: Light Curve

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: RATE				
1	TIME	Float (64-bit)	s	Bin mid-point since TIMEZERO+TIMEZERF
2	RATE	Float (64-bit)	counts/sec	Count rate
3	ERROR	Float (64-bit)	counts/sec	Error in count rate
4	FRAC_EXP	Float (64-bit)	-	Fractional overlap of bin with GTI
Extension 2: GTI				
1	TSTART	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI start time
2	TSTOP	Float (64-bit)	S	UTC (s since POLIX epoch) of GTI stop time

Table 11: Description of L2 data product: Spectrum

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: Spectrum				
1	CHANNEL	Unsigned integer (16-bit)	-	Pulse-invariant channel
2	COUNTS	Float (64-bit)	s-	Counts in the channel
3	STAT ERR	Float (64-bit)	-	Statistical error in the channel

The structure and contents of the light curve and spectrum are identical for source, background, and Earth occultation observations of event file and source covering processed event file. Same applicable for count data light curve of all three source, background, and Earth occultation observations. Therefore, the description provided in this table is applicable to all three cases. The second extension in each file contains the GTI (Good Time Intervals) corresponding to the respective source, background, or Earth occultation observation.

Table 12: Description of L2 data product: Exp_Azimuth_Roll_L2.fits

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: RollAzimuthal				
1	Roll_AZ_ANG	Float (64-bit)		Azimuthal angle in Degrees
2	DET1_Exposure	Float (64-bit)		Exposure time for each angle in sec for Det1
3	DET2_Exposure	Float (64-bit)		Exposure time for each angle in sec for Det2
4	DET3_Exposure	Float (64-bit)		Exposure time for each angle in sec for Det3
5	DET4_Exposure	Float (64-bit)		Exposure time for each angle in sec for Det4

Table 13: Description of L2 data product: Exp_Azimuth_Yaw_L2.fits

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: YawAzimuthal				
1	Yaw_AZ_ANG	Float (64-bit)		Azimuthal angle in Degrees
2	DET1_Exposure	Float (64-bit)		Exposure time for each angle in sec for Det1
3	DET2_Exposure	Float (64-bit)		Exposure time for each angle in sec for Det2
4	DET3_Exposure	Float (64-bit)		Exposure time for each angle in sec for Det3
5	DET4_Exposure	Float (64-bit)		Exposure time for each angle in sec for Det4

Table 14: Description of L2 data product: Src_Azimuth_Roll_L2.fits

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: RollAzimuthal				
1	Roll_AZ_ANG	Float (64-bit)	-	Azimuthal angle in Degrees
2	ANODE_COUNTS	Unsigned integer (16-bit)	-	Count of 48 anode cells

Table 15: Description of L2 data product: Src_Azimuth_Yaw_L2.fits

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: YawAzimuthal				
1	Yaw_AZ_ANG	Float (64-bit)	-	Azimuthal angle in Degrees
2	ANODE_COUNTS	Unsigned integer (16-bit)	-	Count of 48 anode cells

Table 16: Description of L2 data product: EnergyRes_Src_Azimuth_Roll_L2.fits

<i>Sl.no</i>		<i>Type and Structure</i>	<i>Unit</i>	<i>Description</i>
Extension 1: Primary				
1		Int2 (48 × 8192 × 360)	-	Energy-resolved, anode-wise azimuthal modulation curves of detected photons
Axis description of Primary Array (48 × 8192 × 360)				
Axis	Size	Type	Unit	Description
NAXIS1	48	Int2	-	Number of anodes (each index corresponds to one of 48 anodes)

NAXIS2	8192	Int2	-	Energy channels (PHA bins spanning full detector energy range)
NAXIS3	360	Int2	-	Azimuthal angle bins (1° per bin)

Table 17: Description of L2 data product: WeightedRoll_L2.fits

<i>Sl.no</i>	<i>Column name</i>	<i>Type</i>	<i>Unit</i>	<i>Description</i>
Extension 1: ROLLSHIFTWEIGHTEDADD				
1	Roll_AZ_ANG	Float (64-bit)	-	Azimuthal angle in Degrees
2	TOTAL_COUNTRATE	Float (64-bit)	-	Count rate for each azimuthal angle
3	ERROR	Float (64-bit)	-	Error on count rate